Batch inference: which models are worth paying for?

Benchmark-weighted quality against cost for eight common batch workloads, 71 OpenRouter `:batch` models

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Written by	Claude Opus 5.5 (claude-opus-5-5), for Daniel Rosehill. Every figure, table and finding in this report was generated by Claude from the scripts and data listed in section 6. Not reviewed by a human before publication.
Data as of	Prices: OpenRouter `/api/v1/models`, 2026-10-31. LMArena text: published 2026-10-28. LMArena document: 2026-09-13. Vectara: 2026-10-22.
Repository	`ai-api-costs/OpenRouter-Batch-Notes`, notebook entries 08 (costs) and 09 (this method)

1. Executive summary

- **No benchmark measures batch work directly.** A search on 2026-10-31 found no public leaderboard scoring quality against cost on labelling, extraction or synthetic-data jobs. Everything here is a proxy built from human-preference (LMArena) and summary-faithfulness (Vectara) scores. A labelled pilot of your own (cost per correct output) beats any number in this report.
- **Two cheap models carry the bottom of every frontier.** GLM-5.3 Flash and DeepSeek V4.1 Flash score around the median batch model at US$0.08–0.95 per 1,000 requests, depending on workload. GLM-5.3 Flash's score is the most trustworthy of the cheap ones: LMArena lists it with no reasoning-effort tag, so it should hold with reasoning off.
- **For synthetic data (prompt pairs), quality is expensive at the top.** Over GLM-5.3 Flash, Gemini 3.8 Flash costs about 10× for +0.04 quality, and Claude Opus 5.5 about 50× for +0.10. A cheap generator plus a filtering pass is probably the better spend. That is inferred, not tested.
- **Gemini 3.8 Flash is the "good enough" pick** (★: the cheapest model within 90% of the best) for `classify`, `extract`, `translate` and `synth`. GLM-5.3 Flash is the pick for `judge` and `reason`, and DeepSeek V4.1 Flash for `summarise`.
- **The gpt-oss models are the cheapest but sit well below median** on every LMArena category.
- **Main caveat: reasoning variants.** For about 39% of models, the only LMArena entry is a high- or max-effort variant, while the costs assume reasoning off. Those rows are marked † and probably overstate quality at the quoted price.
- **Coverage gaps.** LMArena scores 58 of 71 models. None of the nine `-pro` ids or `gpt-6.1-terra` is scored. `doc_qa` has only 24 models and none from the cheapest group, so treat that table as a gap.

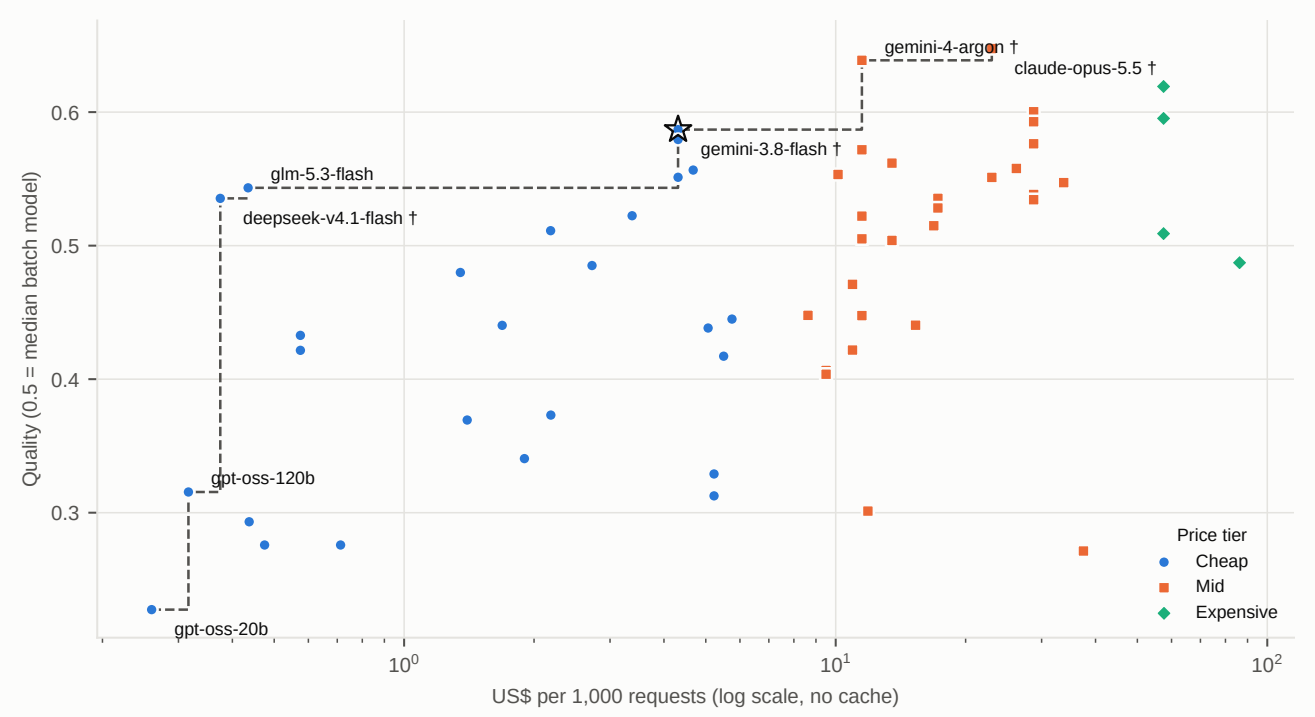


Figure 1: Synthetic data generation (`synth`, 1,500 in / 2,000 out tokens per request). Each mark is one batch model; the dashed step is the Pareto frontier (no model is both cheaper and better). Labelled models are on the frontier; ★ is the good-enough pick; † = score from a reasoning variant.

2. Question and approach

Entry 08 costed each workload on every batch model. This report adds a quality estimate per workload and asks which models are worth their price. Signals were kept small on purpose: two open sources, one to three signals per workload, chosen from benchmarks that already score most current models. Better-fitting benchmarks with thin or stale coverage were left out (section 5).

2.1. Sources

Source	What it measures	Batch ids scored	Access
LMArena text	Human pairwise preference (Elo) by category: instruction following, creative writing, hard prompts, expert, non-English, longer query, maths, coding	58 / 71	HF `lmarena-ai/leaderboard-dataset`, CC-BY-4.0
LMArena document	Preference on questions about an uploaded document	24 / 71	same dataset, `document`
Vectara	Factual consistency rate when summarising (HHEM)	33 / 71	GitHub `vectara/hallucination-leaderboard`

Artificial Analysis was not used. It has better-fitting evals (IFBench, AA-LCR long context, GPQA, HLE, tau2), but its API needs a free key and its terms forbid scraping. No key existed in Infisical or 1Password on 2026-10-31. Adding one is the single biggest upgrade available (section 5).

2.2. Signals and weights

Workload	Tokens in / out	Signals (weight)
`classify`	1,600 / 20	instruction_following 1.0
`extract`	4,000 / 500	instruction_following 0.6, hard_prompts 0.4
`summarise`	8,000 / 600	Vectara factual consistency 0.4, document arena 0.4, longer_query 0.2
`doc_qa`	20,000 / 300	document arena 0.7, longer_query 0.3
`judge`	3,000 / 250	expert 0.5, hard_prompts 0.5
`translate`	2,000 / 900	non_english 1.0. There is no Hebrew category, so Hebrew is not measured.
`synth`	1,500 / 2,000	creative_writing 0.5, instruction_following 0.5
`reason`	1,000 / 5,000	math 0.5, coding 0.5 (highest-effort variant)

The weights are judgement calls, not fitted to measured task accuracy.

2.3. Scoring

Elo signal	$q = 1/\left(1 + 10^{(m-r)/400}\right)$, with m the median rating among batch models. q is the chance of beating the median batch model, so 0.5 = median.
Vectara	$q = 0.1 + 0.8(f - f_{\min})/(f_{\max} - f_{\min})$ over batch models with a score.
Quality	Weighted mean of the signals a model has, with weights renormalised over those present. Left blank if the present weights sum to less than 0.5.
Pareto frontier	Sorted by cost (no cache), the models that beat every cheaper model on quality.
★ good-enough pick	The cheapest model with at least 90% of the best quality in that workload.
Variant rule	`reason` uses the highest-effort LMArena variant; every other workload the lowest-effort one, to match entry 08′s no-reasoning costs. † marks rows where only a medium-or-higher-effort variant existed.

3. Findings by workload

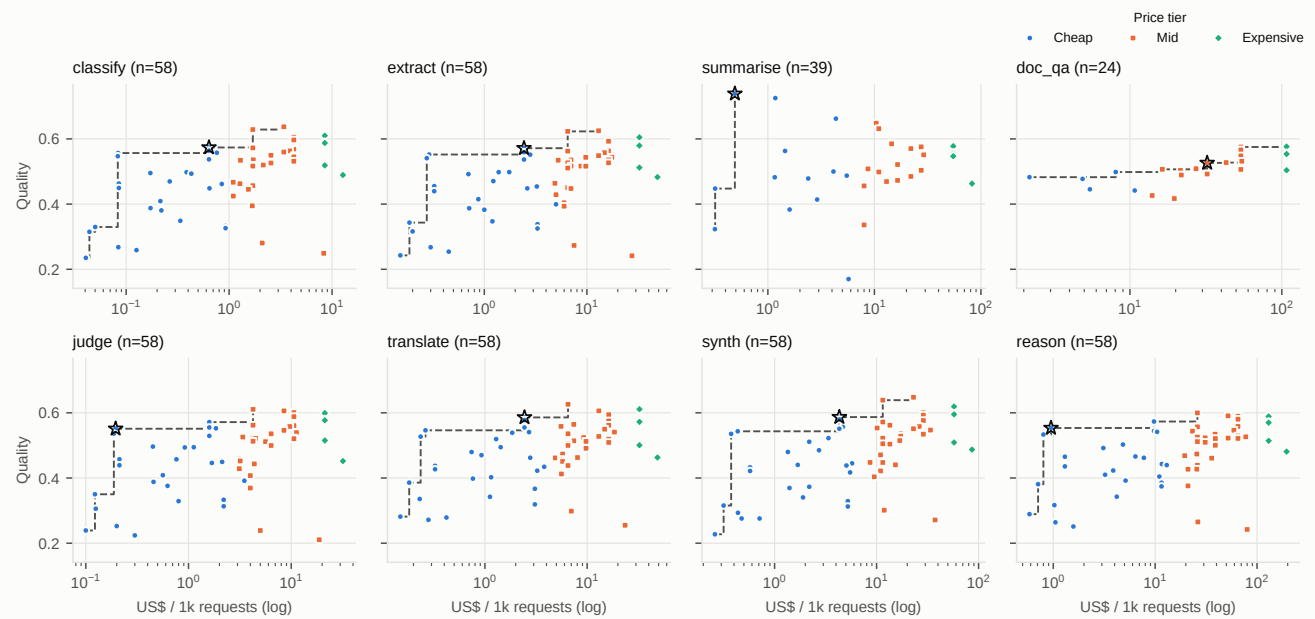


Figure 2: Quality against cost for all eight workloads (same axes and encoding as the headline chart; dashed = frontier, ⋆ = good-enough pick). The frontier shape barely changes between LMArena-scored workloads: the signals are correlated categories of one arena.

The frontier models per workload, cheapest first. Cost is US$ per 1,000 requests without caching (entry 08). Cached costs are in the CSV and only matter for `classify` and `doc_qa` (entry 08).

`classify` · 58 of 71 scored

Model	Tier	Quality	US$/1k	
gpt-oss-20b	Cheap	0.235	0.041	
gpt-5-nano	Cheap	0.315	0.044	†
gpt-oss-120b	Cheap	0.33	0.05	
deepseek-v4.1-flash	Cheap	0.547	0.083	†
glm-5.3-flash	Cheap	0.557	0.084	
gemini-3.8-flash	Cheap	0.574	0.638	⋆†
gemini-4-argon	Mid	0.629	1.7	†
claude-opus-5.5	Mid	0.637	3.4	†

`extract` · 58 of 71 scored

Model	Tier	Quality	US$/1k	
gpt-oss-20b	Cheap	0.243	0.152	
gpt-oss-120b	Cheap	0.343	0.186	
deepseek-v4.1-flash	Cheap	0.541	0.275	†
glm-5.3-flash	Cheap	0.552	0.29	
gemini-3.8-flash	Cheap	0.572	2.438	⋆†
gemini-4-argon	Mid	0.623	6.5	†
claude-opus-5.5	Mid	0.625	13	†

`summarise` · 39 of 71 scored

Model	Tier	Quality	US$/1k	
gpt-oss-120b	Cheap	0.323	0.318	
gpt-5-nano	Cheap	0.448	0.32	†
deepseek-v4.1-flash	Cheap	0.739	0.49	⋆†

`doc_qa` · 24 of 71 scored

Model	Tier	Quality	US$/1k	
gpt-5.6-luna	Cheap	0.483	2.18	†
gemini-3.6-flash	Cheap	0.498	8.063	†
gemini-3.5-flash	Mid	0.507	16.35	†
gpt-5.4	Mid	0.509	27.25	
claude-sonnet-4.6	Mid	0.526	32.25	⋆
gpt-5.6-sol	Mid	0.527	43	†
claude-opus-5	Mid	0.575	53.75	†
claude-fable-5.1	Expensive	0.576	107.5	†

`judge` · 58 of 71 scored

Model	Tier	Quality	US$/1k	
gpt-oss-20b	Cheap	0.239	0.1	
gpt-oss-120b	Cheap	0.35	0.123	
deepseek-v4.1-flash	Cheap	0.536	0.188	†
glm-5.3-flash	Cheap	0.551	0.195	⋆
gemini-3.8-flash	Cheap	0.572	1.594	†
gemini-4-argon	Mid	0.611	4.25	†

`translate` · 58 of 71 scored

Model	Tier	Quality	US$/1k	
gpt-oss-20b	Cheap	0.282	0.149	
gpt-oss-120b	Cheap	0.386	0.182	
deepseek-v4.1-flash	Cheap	0.527	0.235	†
glm-5.3-flash	Cheap	0.546	0.262	
gemini-3.8-flash	Cheap	0.586	2.438	⋆†
gemini-4-argon	Mid	0.626	6.5	†

synth · 58 of 71 scored

Model	Tier	Quality	US$/1k	
gpt-oss-20b	Cheap	0.227	0.26	
gpt-oss-120b	Cheap	0.316	0.316	
deepseek-v4.1-flash	Cheap	0.535	0.375	†
glm-5.3-flash	Cheap	0.543	0.435	
gemini-3.8-flash	Cheap	0.587	4.313	⋆†
gemini-4-argon	Mid	0.639	11.5	†
claude-opus-5.5	Mid	0.648	23	†

reason · 58 of 71 scored

Model	Tier	Quality	US$/1k	
gpt-oss-20b	Cheap	0.289	0.584	
gpt-oss-120b	Cheap	0.381	0.71	
deepseek-v4.1-flash	Cheap	0.533	0.8	
glm-5.3-flash	Cheap	0.554	0.95	⋆
gemini-3.8-flash	Cheap	0.573	9.75	
gemini-4-argon	Mid	0.6	26	

3.1. Notes per workload

- **`classify`, `extract`, `judge`, `translate`, `synth`:** the frontier runs gpt-oss-20b → gpt-oss-120b → DeepSeek V4.1 Flash → GLM-5.3 Flash → Gemini 3.8 Flash → Gemini 4 Argon / Claude Opus 5.5. The jump from GLM-5.3 Flash to Gemini 3.8 Flash is about 8–10× in cost for +0.02 to +0.04 quality.
- **`summarise`:** DeepSeek V4.1 Flash is the ⋆ pick (Vectara factual consistency 95.8%). Only gpt-5.4-nano scores higher (96.9%), at 2.4× the cost. The cheap end rests mostly on Vectara, because those models aren't in the document arena (coverage 0.6).
- **`doc_qa`:** only the 24 models in the document arena are scored, and none is from the cheapest group. Claude Sonnet 4.6 is the ⋆ pick at US$32 per 1,000 requests. Caching cuts this workload's cost by 44–86% (entry 08), which this table does not show.
- **`reason`:** this is the only workload scored on highest-effort variants, so no row is †. GLM-5.3 Flash is the ⋆ pick.

4. Recommendations for synthetic data generation

1. **Pilot GLM-5.3 Flash and DeepSeek V4.1 Flash first** for prompt-pair generation, at about US$0.38–0.44 per 1,000 samples (no cache). Gemini 3.8 Flash (US$4.31) is the step up if the pilot's pass rate is poor.
2. **Measure what the arena can't:** diversity (duplicate and near-duplicate rate across samples), schema validity, and pass rate under your own filter or judge. Cost per *accepted* pair is the number to compare.
3. **Watch hidden reasoning.** DeepSeek V4.1 Flash's arena score is from its Max variant. With reasoning off (entry 03), quality may drop. With it on, output tokens and cost rise.
4. **Caching barely helps here** (6–17% best case, entry 08), because output dominates the bill. Rank candidates by output price.

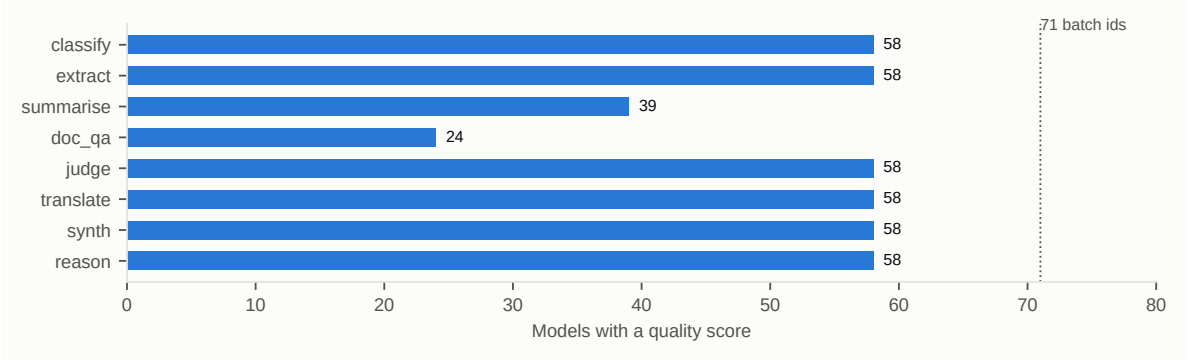


Figure 3: Models with a quality score per workload, out of 71 batch ids. `summarise` and `doc_qa` depend on the smaller document arena and Vectara lists.

5. Limitations

- **Proxies, not task accuracy.** Arena Elo is human preference on chat prompts. A preferred answer is not necessarily a correct label or a valid JSON object.
- **Compressed scale.** All models fall between about 0.2 and 0.65, so differences under about 0.02 are ties.
- **Reasoning mismatch** (†): about 39% of rows, mostly newer models.
- **Hand mapping.** OpenRouter ids were matched to source names by hand (`reference/benchmarks/model-map.json`). Dated snapshots (GPT-4o, o3-mini, older Claude) may not be exactly the one OpenRouter serves.
- **Hebrew** is not measured by any source used.
- **Rejected as stale or without a leaderboard:** JSONSchemaBench (2025-03), StructEval (2025-09), RewardBench 2 (2025-12), AgoraBench (2024-12), NoveltyBench (none) and LiveBench categories (2025-04).
- **Next upgrade:** an Artificial Analysis key. AA-LCR would replace the document arena for `doc_qa`, and IFBench would join instruction following for `classify`, `extract` and `synth`.

6. Reproducibility

Scores	`uv run --no-project --with pandas --with pyarrow python -I scripts/benchmark_scores.py --fetch /tmp/bench-src reference/workloads/2026-10-31_workload-costs.csv reference/benchmarks/2026-10-31`
Charts	`uv run --no-project --with pandas --with matplotlib python -I scripts/report_charts.py reference/benchmarks/2026-10-31_value.csv reports/2026-10-31_benchmark-value/fig`
PDF	`typst compile --root . reports/2026-10-31_benchmark-value/report.typ reports/2026-10-31_benchmark-value/2026-10-31_benchmark-value-report.pdf`
Weights	`reference/benchmarks/workload-benchmarks.json`
Costs	`reference/workloads/2026-10-31_workload-costs.csv` (entry 08)
Attribution	LMArena (CC-BY-4.0), Vectara hallucination leaderboard. Prices: OpenRouter.

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